

TECHBIZ SECURITY

CYBER SECURITY INTERNSHIP REPORT

Title

Threat Intelligence & Indicators of Compromise (IOCs)

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Threat Intelligence & IOCs

1. INTRODUCTION

This report documents Week 5 – Task 14 (SOC) of the Cyber Security Internship at TechBiz Security. The task introduced the basics of Threat Intelligence and Indicators of Compromise (IOCs), and required investigating suspicious network and file indicators with publicly available OSINT and threat-intelligence platforms.

The work covered the TryHackMe Threat Intelligence Tools room, an email investigation task, a ThreatFox / Abuse.ch IOC record, an AbuseIPDB lookup of an IP address, and a VirusTotal lookup of a file hash. For each artifact, the evidence was interpreted with the aim of separating reputation data, threat-intelligence association, and strong malicious indicators, as expected of a SOC analyst. All statements in this report are limited to what is visible in the captured screenshots.

2. OBJECTIVES

1. Understand IOC types.
2. Understand the purpose of Threat Intelligence.
3. Complete the TryHackMe Threat Intelligence Tools room.
4. Investigate an IP address using AbuseIPDB.
5. Investigate threat-intelligence information using ThreatFox/Abuse.ch.
6. Investigate a file hash using VirusTotal.
7. Interpret the collected evidence.
8. Document findings professionally.

3. IOC TYPES

Threat Intelligence is information about threats, threat infrastructure, and malicious tooling that helps defenders decide what to look for and how to prioritise alerts. IOCs are the observable artifacts that link activity to a known or suspected threat. The four IOC types covered in this task are described below.

3.1 IP Address

What it represents: the numeric network address of a host or server (IPv4 in this task). **SOC use:** analysts check an IP address against reputation and threat-intelligence sources and review its ownership details such as ISP, ASN, and country. **Detection and investigation:** log entries, alerts, and network connections can be compared with known suspicious infrastructure, and events involving the same address can be linked together.

3.2 Domain

What it represents: a human-readable name that resolves to one or more IP addresses. **SOC use:** analysts review hosting and registration details and check whether the domain appears in threat-intelligence data. **Detection and investigation:** DNS, proxy, and email logs can be searched for hosts that contacted a domain of interest.

3.3 URL

What it represents: the full address of a specific web resource, including protocol, domain, and path. **SOC use:** analysts examine URLs found in emails, alerts, and web logs, often with scanning services, to understand where a link leads. **Detection and investigation:** a URL is more specific than a domain, so it can point to an individual page or file location referenced in an alert or message.

3.4 File Hash

What it represents: a fixed-length value calculated from the contents of a file (for example SHA-256), which acts as a fingerprint for that exact file. **SOC use:** analysts search the hash in services such as VirusTotal to see whether the file is known and how security vendors classify it, without needing to open the file. **Detection and investigation:** the same hash can be searched across endpoint and log data to find other copies of the file.

Scope note: in the provided evidence, IP address and file hash IOCs were investigated. A domain name (waicore.com) appears as an attribute in the AbuseIPDB result, but a separate domain or URL investigation is not demonstrated in the provided evidence.

4. TOOLS USED

Only the tools and platforms demonstrated in the screenshots are listed. The screenshots were captured in a Firefox browser within a Kali Linux environment (the VirtualBox window title is visible in Figure 2).

Tool / Platform	Purpose in this investigation
TryHackMe – Threat Intelligence Tools	Guided room covering OSINT tools for threat assessments and investigations; also hosted the email investigation questions.
ThreatFox / Abuse.ch	Threat-intelligence platform used to view the IOC record for 212.192.246.30:5555, including malware family, threat type, and confidence level.
AbuseIPDB	IP reputation database used to check report count, Confidence of Abuse, and ownership details for 212.192.246.30.
VirusTotal	Multi-vendor analysis service used to look up the file hash for blOgger.exe and view vendor detections and threat labels.
Mozilla Thunderbird	Email client visible in the TryHackMe email investigation, displaying the email whose sender, recipient, and originating IP were examined.

Browser tabs for SSLBL, Feodo Tracker, and MalwareBazaar are visible in Figures 3 and 4, but no results from those pages are demonstrated in the provided evidence.

5. TRYHACKME – THREAT INTELLIGENCE TOOLS

The TryHackMe Threat Intelligence Tools room was completed. The screenshot shows the room title “Threat Intelligence Tools”, with a description stating that the room explores different OSINT tools used to conduct security threat assessments and investigations. The completion banner reads “Room completed: 100%”. The following tasks are visible and marked with a completion tick:

- Task 1: Room Outline
- Task 2: Threat Intelligence
- Task 3: UrlScan.io
- Task 4: Abuse.ch
- Task 5: PhishTool

No task answers or additional task results are visible in this screenshot, and tasks below Task 5 are not visible.

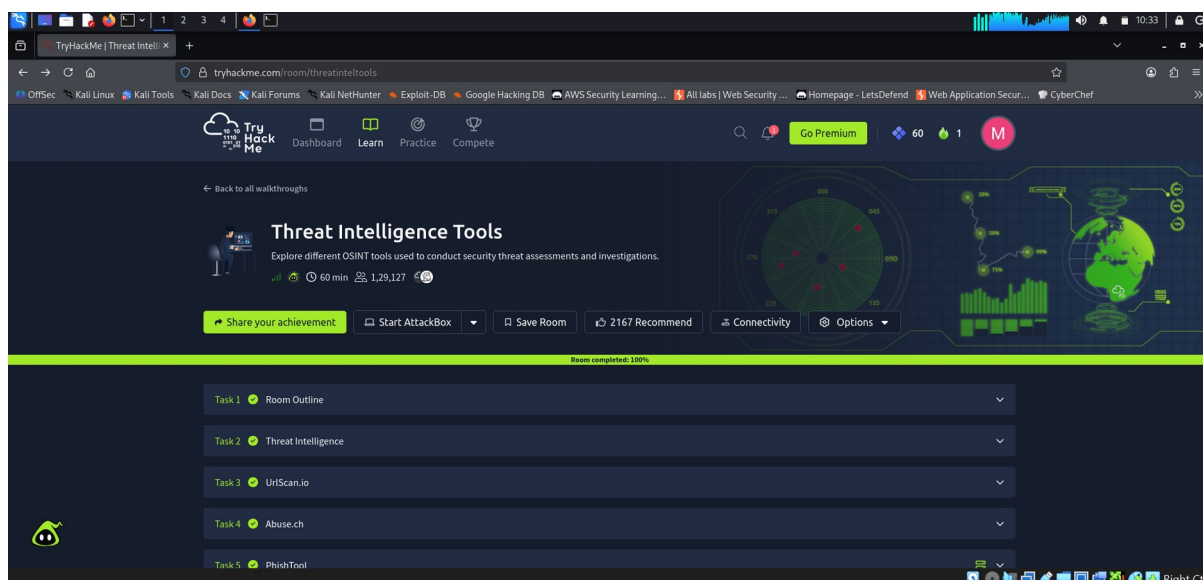


Figure 1: TryHackMe Threat Intelligence Tools room showing 100% completion.

6. THREAT INTELLIGENCE – EMAIL INVESTIGATION

The screenshot shows an email investigation task within the TryHackMe room, with Mozilla Thunderbird displaying the email content. The visible questions and answers are:

Question shown	Answer shown	Status
What is the senders email address?	darkabutta@sc500.whpservers.com	Correct Answer
What is the recipient's email address?	cabbagecare@hotmail.com	Correct Answer
What is the Originating IP address? Defang the IP address.	204[.]193[.]183[.]11	Correct Answer
How many hops did the email go through to get to the recipient?	Not demonstrated in the provided evidence.	Correct Answer

The originating IP is written in defanged format, in which the dots are enclosed in square brackets so that the address is not accidentally clicked or resolved. The answer field for the hops question is not legible in the screenshot, so the hop count is not reported here.

Visible email details in Thunderbird: the sender is displayed as Patrick Cook with the address darkabutta@sc500.whpservers.com, addressed to cabbagecare@hotmail.com, dated 3/29/22, 15:39, with the subject line “You have 5 new message”. A Thunderbird banner states that remote content in the message was blocked. The message body shows “CabbageCare”, the text “You have 5 new message(s) by Patrick Cook”, a “Show message” button, and a LinkedIn app download prompt. The room progress banner in this capture reads 68%, and the final 100% completion is shown in Figure 1.

The screenshot does not include a classification of the email, and the originating IP address was not checked against any threat-intelligence source in the provided evidence.

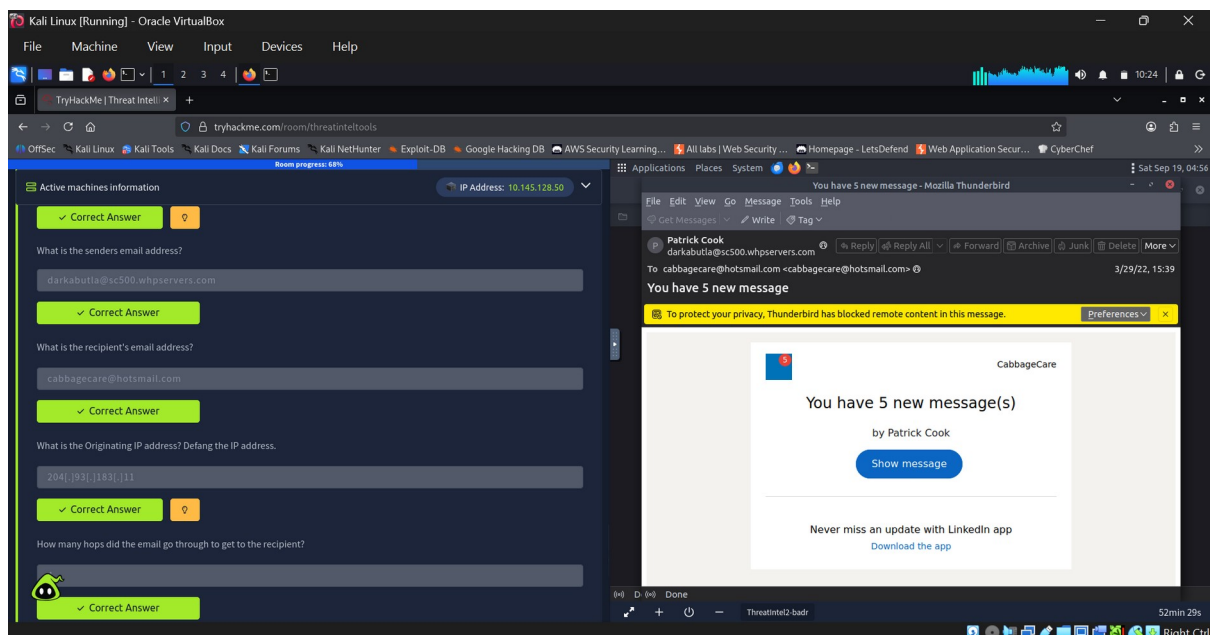


Figure 2: TryHackMe email investigation showing correct answers and email evidence.

7. THREATFOX / ABUSE.CH INVESTIGATION

IOC investigated: 212.192.246.30:5555. In the captured screenshot the top of the record is scrolled out of view, and the browser tab title shows the truncated value “212.192.246.3”; the full IOC value is taken from the task information. The page address shown is threatfox.abuse.ch/ioc/395319/.

ThreatFox field	Value shown
Threat Type	botnet_cc
Malware	Mirai
Malware alias	Katana
Confidence Level	Confidence level is elevated (75%)
Is compromised?	False
ASN	AS44559 ITHOSTLINE
Country	IN
First seen	2022-03-15 07:20:31 UTC
Last seen	2025-08-18 16:22:03 UTC
Tags	Mirai
Reporter	abuse_ch
Reference	A link to a MalwareBazaar sample page (bazaar.abuse.ch)

The reporter comment visible at the bottom of the screenshot reads “Mirai (aka Katana) botnet C2”.

Interpretation. The ThreatFox record associates 212.192.246.30:5555 with Mirai botnet C2 activity and displays an elevated confidence level of 75%. However, the record itself indicates “Is compromised?: False”, so this should be treated as threat-intelligence association rather than proof that a specific endpoint is currently compromised. The Last seen timestamp (2025-08-18) should also be taken into account when judging whether the indicator is still active. The referenced MalwareBazaar sample was not downloaded or analyzed, and no analysis of it is demonstrated in the provided evidence.

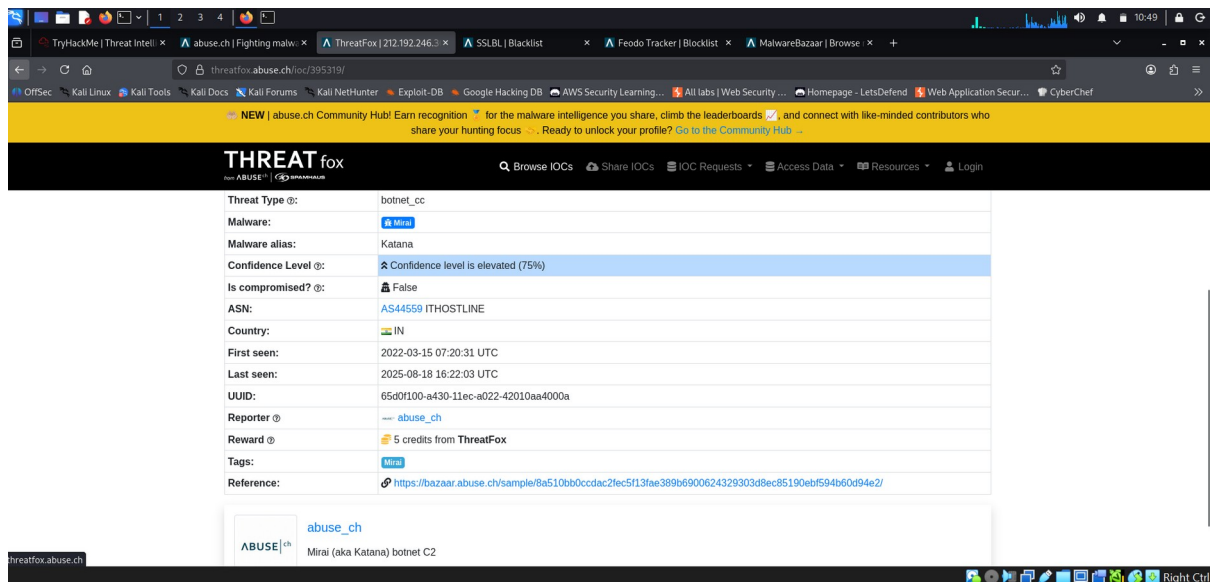


Figure 3: ThreatFox IOC record showing Mirai classification and threat-intelligence details.

8. ABUSEIPDB IP INVESTIGATION

IP address investigated: 212.192.246.30. The AbuseIPDB page states that the IP was found in the database.

AbuseIPDB field	Value shown
Reports	This IP was reported 236 times
Confidence of Abuse	0%
ISP	WAlcore Ltd
Usage Type	Data Center/Web Hosting/Transit
ASN	AS213887
Domain Name	waicore.com
Country	Germany
City	Frankfurt am Main, Hesse

Interpretation. AbuseIPDB shows that the IP address has been reported 236 times, but the current displayed Confidence of Abuse is 0%. Therefore, the AbuseIPDB result alone does not provide sufficient evidence to classify the IP as malicious. The report categories and the date of the most recent report are not visible, so they are not demonstrated in the provided evidence.

Comparison with ThreatFox (Section 7). The two sources give different views of the same address. ThreatFox links 212.192.246.30:5555 to Mirai botnet C2 with an elevated confidence level of 75%, while AbuseIPDB reports 236 reports alongside a 0% Confidence of Abuse. The ownership details also differ: ThreatFox shows AS44559 ITHOSTLINE with country IN, and AbuseIPDB shows AS213887, WAlcore Ltd, and Germany. AbuseIPDB states that its ISP, usage type, and location data is provided by IPinfo and updated weekly. The reason for the difference is not determined from the evidence. The AbuseIPDB result does not confirm the ThreatFox association, and it does not remove it either.

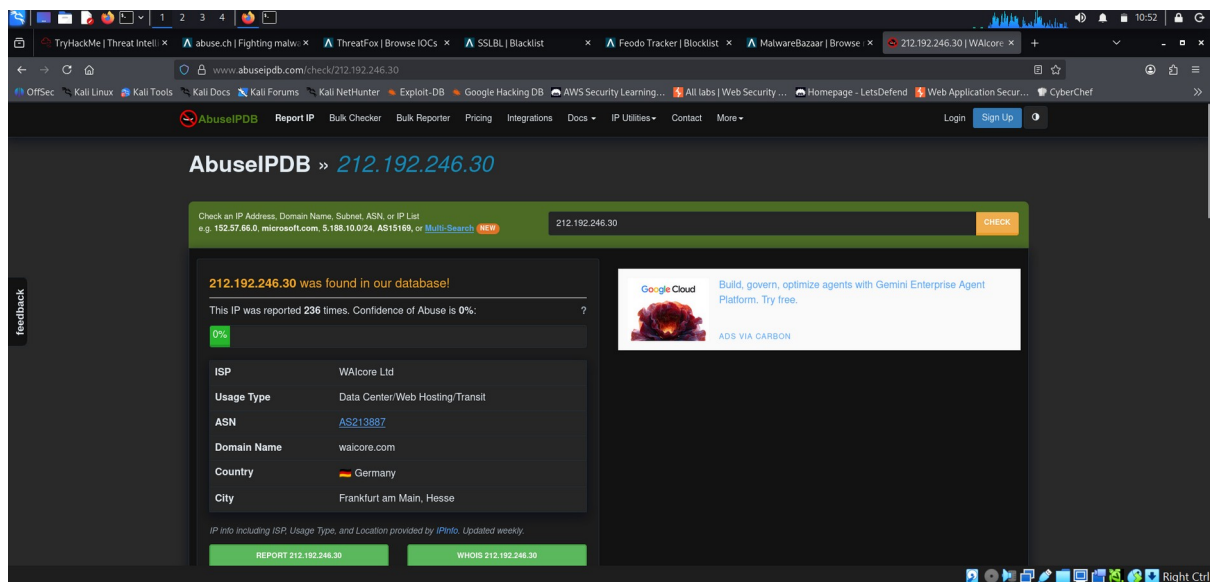


Figure 4: AbuseIPDB investigation of IP address 212.192.246.30.

9. VIRUSTOTAL FILE HASH INVESTIGATION

The file hash was searched in VirusTotal. The result shows the following information:

VirusTotal field	Value shown
File name	bl0gger.exe
File hash	2672b6688d7b32a90f9153d2ff607d6801e6cbde61f509ed36d0450745998d58 (64 hexadecimal characters, the length of a SHA-256 value)
Detection result	58/66 security vendors flagged this file as malicious
Popular threat label	trojan.grafter/blackmoon
Threat categories	trojan, hacktool, spyware
Family labels	grafter, blackmoon, flystudio
File type	EXE
Size	2.57 MB
Last Analysis Date	2 days ago (as displayed; the exact date is not shown)
Community Score	-17 (Community tab shows 22)
Tags displayed	peexe, spreader, overlay, detect-debug-environment

The first ten vendor rows are visible in the screenshot, all showing a detection label:

Vendor	Label shown	Vendor	Label shown
AhnLab-V3	Trojan/Win32.Gen	Alibaba	Malware:Win32/km_249dd1.None
AliCloud	Ransomware:Win/Zusy.Gen	ALYac	Gen:Variant.Application.Graftor.484079
Antiy-AVL	HackTool[VirTool]/Win32.Agent	Arcabit	Trojan.Application.Graftor.D762EF
Arctic Wolf	Unsafe	Avast	Win32:Malware-gen
AVG	Win32:Malware-gen	Avira (no cloud)	TR/W32.Malware

Interpretation. A detection by 58 of 66 security vendors is strong evidence that the analyzed file is malicious or associated with malware, because a large majority of the displayed vendors flagged it. The popular threat label and the trojan, hacktool, and spyware categories are consistent with this. Vendor labels differ from one another, which is normal for multi-vendor results. The evidence shows a hash

lookup only. The displayed tags (such as spreader and detect-debug-environment) were not verified through the Behavior tab, and no file download, execution, or behavioral analysis is demonstrated. The community comments are not visible, so their content is not demonstrated in the provided evidence.

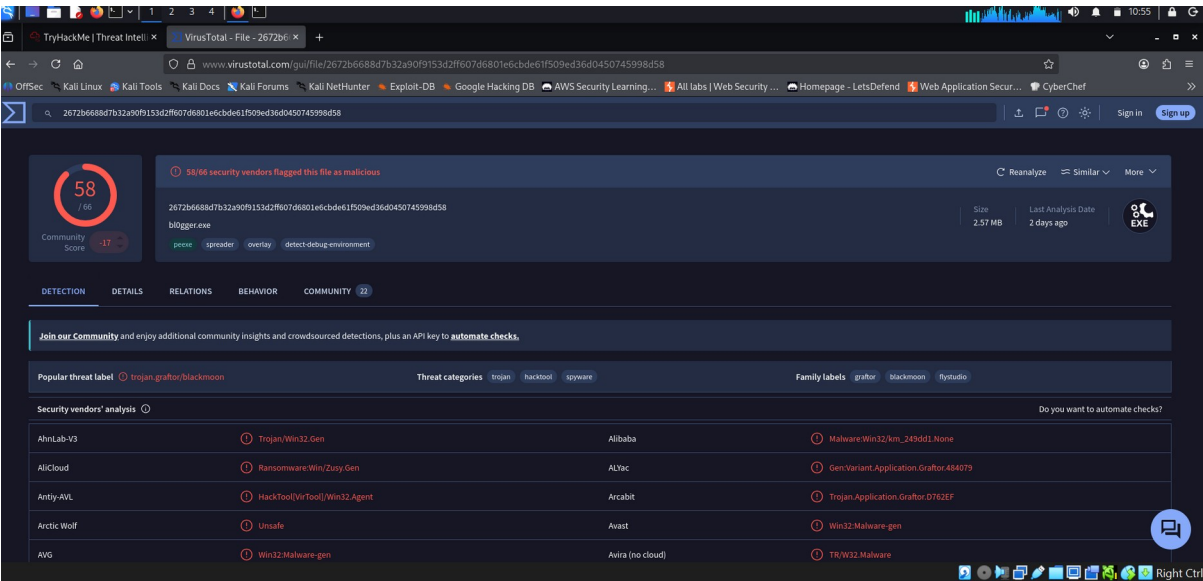


Figure 5: VirusTotal analysis showing 58/66 security vendors flagging blogger.exe as malicious.

10. IOC INVESTIGATION SUMMARY

IOC / Artifact	Source	Observed Finding	Interpretation
212.192.246.30:5555	ThreatFox	Mirai, botnet_cc, elevated confidence 75%	Threat-intelligence association with Mirai botnet C2; not proof of current compromise
212.192.246.30	AbuseIPDB	236 reports, 0% Confidence of Abuse	Report history exists, but displayed confidence does not independently support a malicious verdict
2672b6688d7b32a90f9153d2ff607d6801e6cbde61f509ed36d0450745998d58	VirusTotal	58/66 vendors flagged blogger.exe	Strong malicious indicator based on the displayed multi-vendor detections

11. CORRELATION AND ANALYSIS

The IP address 212.192.246.30 was investigated through both ThreatFox and AbuseIPDB. ThreatFox provides a threat-intelligence association with Mirai botnet C2 and displays an elevated confidence level of 75%. AbuseIPDB shows 236 reports but a displayed Confidence of Abuse of 0%. Differences in data source, scoring method, update time, and the fact that the ThreatFox record refers to port 5555 while AbuseIPDB reports at IP level may all contribute, but the evidence does not show which factor applies.

These results demonstrate why SOC analysts should correlate multiple intelligence sources instead of relying on a single reputation score. A 0% Confidence of Abuse does not cancel a threat-intelligence association, and a report count alone does not establish a malicious verdict.

The file hash investigated through VirusTotal has 58/66 vendor detections, which is substantially stronger evidence of malicious classification than the AbuseIPDB reputation result alone. It should still be read as a point-in-time result, since VirusTotal displays a last analysis date.

Evidence should therefore be interpreted according to its source, timestamp, confidence, and context. The provided evidence contains no internal logs, alerts, or endpoint telemetry, so no conclusion is drawn about the impact on any specific system or network.

12. VERDICT

12.1 IP: 212.192.246.30:5555

212.192.246.30:5555 is threat-associated according to the ThreatFox evidence, where it is linked to Mirai botnet C2 with an elevated confidence level of 75%. However, AbuseIPDB displays a 0% Confidence of Abuse despite 236 reports. Therefore, the evidence supports treating the IOC as suspicious/threat-associated and requiring further investigation, rather than claiming confirmed current compromise.

12.2 File: bl0gger.exe

The investigated bl0gger.exe hash has strong malicious indicators in VirusTotal, with 58 of 66 security vendors flagging the file. The displayed threat labels also include trojan-related classifications. Based on the provided VirusTotal evidence, the file should be treated as malicious.

13. SOC RELEVANCE

- IOC enrichment: an IP address or hash from an alert can be enriched with ThreatFox, AbuseIPDB, and VirusTotal to add context before a decision is made.
- Alert investigation: the same checks help an analyst judge whether an alert involving an external address or file needs escalation.
- Threat hunting: confirmed or suspected indicators such as an IP:port pair or file hash can be searched for in logs and endpoint data.
- IP reputation checking: AbuseIPDB provides report counts, Confidence of Abuse, and ownership details, which must be read together and not as a single verdict.
- Malware and hash investigation: VirusTotal shows how many vendors flag a file and which labels they use, without opening the file.
- Correlation of multiple intelligence sources: differing results, as seen for 212.192.246.30, show why more than one source is needed.
- Evidence-based incident triage: verdicts are worded according to the strength of the evidence, for example suspicious/threat-associated versus malicious.
- Identifying potentially malicious infrastructure and files: botnet C2 associations and high vendor detection counts help highlight infrastructure and files that deserve priority.

14. KEY LEARNINGS

- Understanding different IOC types.
- Using OSINT-based threat-intelligence platforms.
- Investigating IP reputation.
- Investigating malware-related IOCs.
- Understanding the difference between reports, confidence scores, and threat-intelligence classifications.
- Using multiple sources to improve investigation accuracy.
- Documenting SOC investigation findings with screenshots and evidence.

15. CONCLUSION

This task provided practical exposure to Threat Intelligence and IOC investigation through TryHackMe, ThreatFox/Abuse.ch, AbuseIPDB, and VirusTotal. The investigation demonstrated the importance of

correlating multiple intelligence sources and interpreting evidence carefully before reaching a verdict. The IP indicator was assessed as suspicious/threat-associated and requiring further investigation, while the file hash was assessed as malicious based on strong multi-vendor detections.

16. EVIDENCE INDEX

No.	Screenshot filename	Used in
1	THM_Threat_Intelligence - Email_Investigation.png	Section 6, Figure 2
2	THM_Threat_Intelligence_Tools - 100%_Completion.png	Section 5, Figure 1
3	ThreatFox_Mirai_IP_Investigation.png	Section 7, Figure 3
4	AbuseIPDB_212.192.246.30_IP_Investigation.png	Section 8, Figure 4
5	VirusTotal_File_Hash_Investigation.png	Section 9, Figure 5